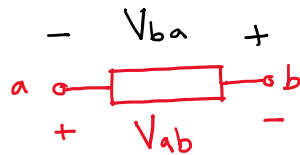
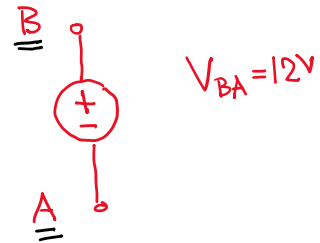
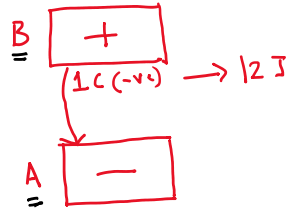


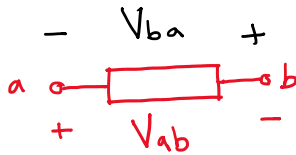
$$V = 12V \Rightarrow V_{BA} = 12V$$



$$V_{ab} = -V_{ba}$$

$$V_{ab} = 10V$$

$$V_{ba} = -V_{ab} = -10V$$



$V_{ab} = -9V$, If you take $-1C$, from

Position a to position b , you have to spend -9 Joule
 $= 9V$ meaning $9J$ is

$V_{ba} = 9V$. $b \uparrow$ Potential, $a \downarrow$ Potential

Active Elements: That supplies electrical energy

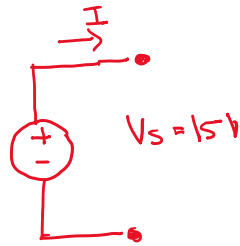
- Voltage Source
- Current Source

Passive elements: Absorbs energy

- Resistors, □ Inductor □ Capacitor

- Independent sources $\rightarrow$ $\begin{cases} \text{Voltage Source} \\ \text{Current Source} \end{cases}$
- Dependent sources

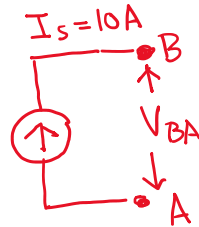
II Dependent sources



Value of the source is independent of the current flowing through it

$$V_s = 15V$$

$$I = 1A, 100A$$

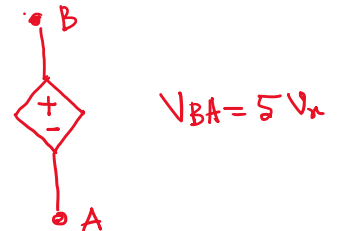


The value of the current is independent of voltage across the source

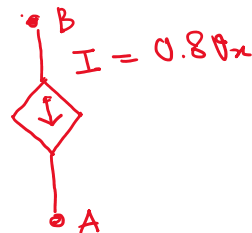
$$V_{BA} = 2V, 10V, 100V$$

Dependent sources: Value of the voltage or current depends on some external value of voltage or current in the circuit

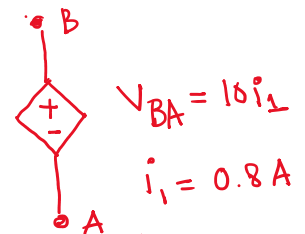
1. Voltage controlled voltage source:



2. Voltage Controlled current source



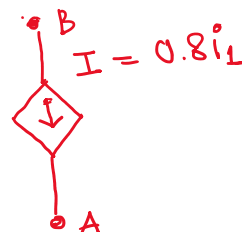
3. Current Controlled voltage source



$$i_1 = 0.8A$$

$$V_{BA} = 10 \times 0.8 = 8V$$

4. Current Controlled current source

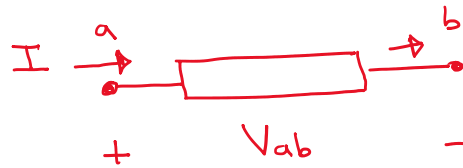


$$i_1 = 10A$$

$$I = 0.8 \times 10 = 8A$$

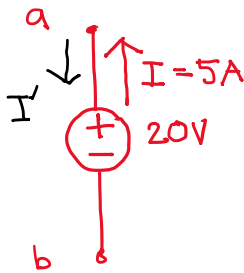


$$I = 0.8 \times 10 = 8A$$



Current enters into the +ve terminal

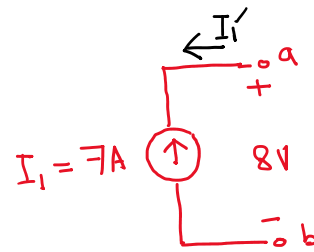
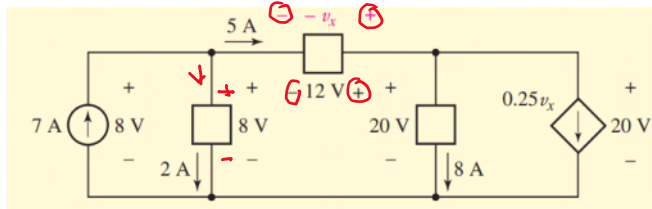
$P = V_{ab} \times I \rightarrow (+ve)$ Absorbs power (Passive elements)
 $(-ve) \rightarrow$ delivering Power (Active elements)



$V_{ab} = 20V$, Assign a current I' so that it enters into the positive terminal

$$P = V_{ab} \times I' = V_{ab} \times (-I) = 20 \times (-5) = -100W$$

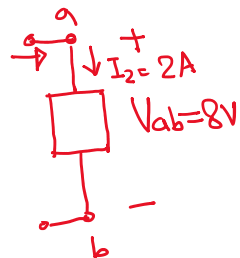
delivering



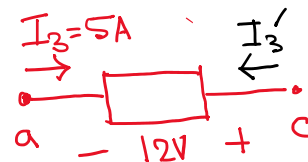
$$V_{ab} = 8V$$

$$I_1' = -I_1 = -7A$$

$$P_1 = V_{ab} \times I_1' = V_{ab} \times (-I_1) = -8 \times 7 = -56W \quad (-ve) \rightarrow \text{delivering Supplying}$$

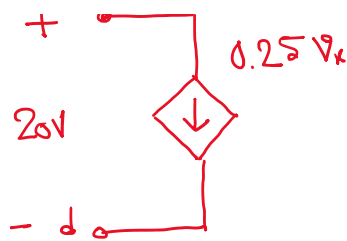


$$P_2 = V_{ab} \times I_2 = 8 \times 2 = 16W$$



$$P_3 = V_{ca} \times I_3' = 12 \times (-I_3) = 12 \times (-5) = -60W \quad (\text{deliver})$$





$$-V_x = 12V$$

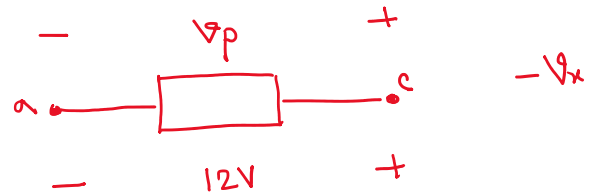
$$\Rightarrow V_x = -12V$$

$$I_S = 0.25 V_x = -0.25 \times 12A = -3A$$

$$P_S = V_C \times I_S = 20 \times (-3) = -60W \text{ (delivering)}$$

total power delivered = total power absorbed

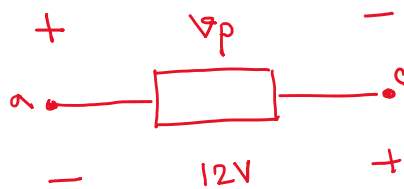
$$V_p = -V_x$$



$$V_p = -V_x$$

$$V_{ca} = 12V, \quad V_{ca} = V_p = -V_x$$

$$-V_x = 12V, \Rightarrow V_x = -12V$$



$$V_{ca} = 12V,$$

$$V_{ac} = V_p$$

$$V_{ca} = -V_{ac}$$

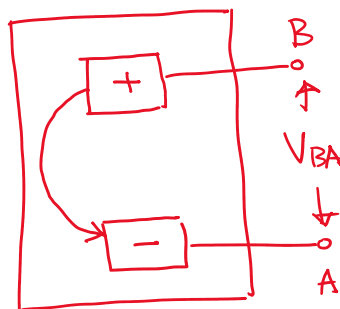
$$\Rightarrow 12 = -V_p$$

$$\Rightarrow V_p = -12V$$

$$\Rightarrow -V_x = -12V$$

$$\Rightarrow V_x = 12V$$

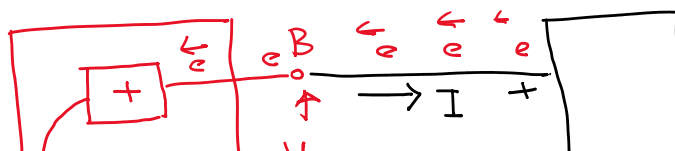
EMF $\rightarrow$ through separation of charges



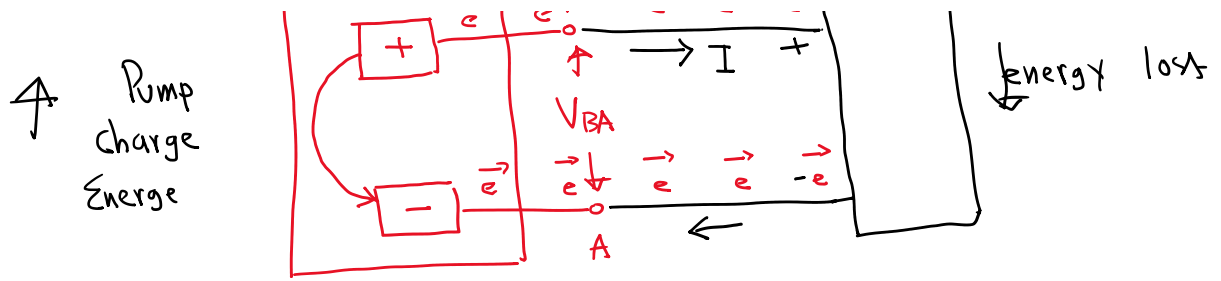
No flow of charges: current ~~Zero~~ Zero

A

Pump



energy loss



1st Pump charge (gains energy) [Battery]

↓
delivers energy to the passive elements
↓
Passive elements absorbs their ~~energy~~ energy